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BIO & RESEARCH INTERESTS

I am a Staff Researcher at the AI Center of Samsung Electronics, specializing in computer vision, deep learning, and machine learning. My primary focus is on Vision-Language-Action (VLA) models for embodied AI and robotics, with the goal of building AI systems that can robustly perceive, reason, and act in complex, real-world environments.

I received my Ph.D. from Seoul National University, Graduate School of Convergence Science and Technology, under the supervision of Prof. Nojun Kwak, and was affiliated with the Machine Intelligence and Pattern Analysis Lab (MIPAL).

My past and ongoing research spans domain adaptation, domain generalization, continual/test-time adaptation, knowledge distillation, object detection, and semantic segmentation. Recently, I have been particularly interested in developing methods to make AI models more adaptive and generalizable to unseen scenarios.

Currently, my main research direction is advancing scalable and generalizable VLA models that integrate vision, language, and action generation for robotic control. I have hands-on experience with leading frameworks such as OpenVLA and Physical Intelligence (π_0). Furthermore, I also have experience with using NVIDIA Isaac GR00T N1.5 for control and Cosmos for world modeling, integrating them to enhance robotic manipulation and decision-making capabilities.

My long-term vision is to enable embodied agents that can learn continuously from diverse environments, follow natural language instructions, and perform complex physical tasks with minimal supervision.

EDUCATION

Seoul National University (SNU) Seoul, Korea Graduate School of Convergence Science and Technology (GSCST) Department of Intelligence and Information <i>Ph.D. in Engineering</i>	Mar. 2019 - Feb. 2024
Yonsei University , Seoul, Korea <i>B.S., Creative Technology Management & Computer Science</i>	Mar. 2012 - Feb. 2019
Technical University of Munich , Germany <i>Exchange Student Program, Informatics(Computer Science)</i>	Oct. 2016 - Aug. 2017
SCSC (Samsung Convergence Software Course) , Yonsei, Korea <i>Educational program designed to educate non-SW major students to become SW expert</i>	Sep. 2015 - Feb. 2019

WORK EXPERIENCE

AI center, Samsung Electronics , Staff Researcher	Dec. 2024 - Present
• Spearheaded the development of Vision-Language-Action (VLA) models for autonomous robotics in semiconductor FAB environments, enabling robots to interpret and execute complex commands via multimodal sensory inputs. • Designed and integrated a hierarchical subtask prediction module atop the OpenPi (π_0) architecture, significantly enhancing task success rates (by 17.5%) and model interpretability for complex manipulation sequences. • Orchestrated the end-to-end model lifecycle, ranging from action data acquisition for multi-object manipulation (pick-and-place) to model training and deployment.	

- Established rigorous evaluation protocols, incorporating comprehensive human-in-the-loop assessments to validate model performance and safety in high-precision industrial settings.

SAIT, Samsung Electronics, Staff Researcher

Mar. 2024 - Dec. 2024

- Conducted a comprehensive feasibility study on applying generative AI and foundation models to robotic manipulation, analyzing the limitations of existing SOTA algorithms.
- Benchmarked various open-source VLA frameworks (e.g., OpenVLA, RT-X) to assess their viability for industrial automation tasks, specifically focusing on generalization capabilities.
- Formulated the strategic research roadmap for integrating multimodal AI into semiconductor manufacturing processes, laying the technical groundwork for subsequent development projects.

Qualcomm AI Research Korea, Research Internship

Jan. 2022 - July. 2022

- Developed novel few-shot learning (FSL) algorithms for keyword spotting (KWS), enabling rapid adaptation to new keywords with minimal labeled data samples.
- Enhanced model robustness and computational efficiency in speech recognition tasks by optimizing architectures for data-scarce scenarios.
- Investigated generalization techniques to mitigate overfitting when training with limited supervision, ensuring high performance on unseen target words.

Naver Webtoons Corp., Research Internship

Jun. 2020 - Dec. 2020

- Conducted research on unsupervised domain adaptation (UDA) for semantic segmentation to address the domain shift problem between realistic and webtoon-stylized imagery.
- Developed novel adaptation techniques tailored for diverse webtoon styles, significantly improving model generalization and robustness across heterogeneous artistic domains.
- Mitigated the performance drop caused by domain gaps, achieving enhanced segmentation accuracy on real-world datasets without reliance on pixel-level target annotations.

Republic of Korea Army, Sergeant

Oct. 2013 - Jul. 2015

PUBLICATIONS

Inseop Chung, Kyomin Hwang, Jayeon Yoo, Nojun Kwak. “Mitigating the Bias in the Model for Continual Test-Time Adaptation”, *IEEE Access*, Aug. 2025

Inseop Chung, Kiyoon Yoo, Nojun Kwak. “Open Domain Generalization with a Single Network by Regularization Exploiting Pretrained Features”, *Workshop on Data-centric Machine Learning Research (DMLR) at The Twelfth International Conference on Learning Representations (ICLR 2024)*, May. 2024

Jayeon Yoo, Dongkwan Lee, **Inseop Chung**, Donghyun Kim, Nojun Kwak, “What, How, and When Should Object Detectors Update in Continually Changing Test Domains?”, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR 2024)*, Jun. 2024

Jayeon Yoo, Dongkwan Lee, **Inseop Chung**, Kyomin Hwang, Seungwoo Lee, Donghyun Kim, Nojun Kwak, “지속적으로 변하는 테스트 환경에서 객체 탐지를 위한 온라인 적응 방법”, *36th Workshop on Image Processing and Image Understanding (IPIU 2024)*, Jan. 2024

Jaeyoung Yoo*, Hojun Lee*, Seunghyeon Seo, **Inseop Chung**, Nojun Kwak. “End-to-End Multi-Object Detection with a Regularized Mixture Model”, *Fortieth International Conference on Machine Learning (ICML 2023)*, Jul. 2023

Inseop Chung, Jayeon Yoo, Nojun Kwak. “Exploiting Inter-pixel Correlations in Unsupervised Domain Adaptation for Semantic Segmentation”, *3rd Workshop on Real-World Surveillance: Applications and Challenges at Winter Conference on Applications of Computer Vision 2023 (The best paper award)(RWS @ WACV 2023)* Jan. 2023

DongKi Noh, Chang Ki Sung, Taeyoung Uhm, Wooju Lee, Hyungtae Lim, Jaeseok Choi, Kyuewang Lee, Dasol Hong, Daeho Um, **Inseop Chung**, Hochul Shin, Min-Jung Kim, Hyoung-Rock Kim, Seung-Min Baek, Hyun Myung, “X-MAS: Extremely Large-Scale Multi-Modal Sensor Dataset for Outdoor Surveillance in Real Environments”, *IEEE Robotics and Automation Letters (RA-L)*, Jan. 2023

Jayeon Yoo, **Inseop Chung**, Nojun Kwak, “Unsupervised Domain Adaptation for One-Stage Object Detector Using Offsets to the Bounding Box”, *European Conference on Computer Vision 2022 (ECCV*

2022), Oct. 2022

Byeonggeun Kim, Seunghan Yang, **Inseop Chung**, Simyung Chang, “Dummy Prototypical Networks for Few-Shot Open-Set Keyword Spotting”, *23rd INTERSPEECH Conference*, Sep. 2022.

Seunghan Yang, Byeonggeun Kim, **Inseop Chung**, Simyung Chang, “Personalized Keyword Spotting through Multi-task Learning”, *23rd INTERSPEECH Conference*, Sep. 2022.

Inseop Chung, Daesik Kim, Nojun Kwak. “Maximizing Cosine Similarity Between Spatial Features for Unsupervised Domain Adaptation in Semantic Segmentation”, *Winter Conference on Applications of Computer Vision 2022 (WACV 2022)*, Jan. 2022

Kyuewang Lee*, **Inseop Chung***, Daeho Um, Jaeseok Choi, Yeji Song, Seunghyeon Seo, Nojun Kwak, Jin Young Choi “Multi-modal Object Detection, Tracking, and Action Classification for Unmanned Outdoor Surveillance Robots”, *The 21st International Conference on Control, Automation and Systems (ICCAS 2021)*, Oct. 2021

Jaeyoung Yoo, Hojun Lee*, **Inseop Chung***, Geonseok Seo, Nojun Kwak. “Training Multi-Object Detector by Estimating Bounding Box Distribution for Input Image”, *International Conference on Computer Vision 2021 (ICCV 2021)*, Oct. 2021

Hojun Lee, **Inseop Chung**, Nojun Kwak, “Multi-modal sensor based framework for object detection”, *The 16th Korea Robotics Society Annual Conference (KRoC 2021)*, May. 2021

Jangho Kim, Minsun Hyun, **Inseop Chung**, Nojun Kwak. “Feature Fusion for Online Mutual Knowledge Distillation”, *25th International Conference on Pattern Recognition (ICPR 2020)*, Jan. 2021

Inseop Chung, SeongUk Park, Jangho Kim, Nojun Kwak. “Feature-map-level Online Adversarial Knowledge Distillation”, *Thirty-seventh International Conference on Machine Learning (ICML 2020)*, Jul. 2020

* indicates equal contribution.

PATENTS

Domestic Patents

Device for Unsupervised Domain Adaptation in Semantic Segmentation Exploiting Inter-pixel Correlations and Driving Method Thereof

patent application number : 1020220035170

patent application date : 2022.03.22

Device for Regression Scale-aware Cross-domain Object Detection and Driving Method Thereof

patent application number : 1020220035183

patent application date : 2022.03.22

International Patents

Method and Apparatus with Online Task Planning

patent application date : 2024.12.09

Method and Apparatus with Micro-Action Determination

patent application number : US19/208393

patent application date : 2025.05.14

Multi-Task Learning For Personalized Keyword Spotting

International Application No. : PCT/US2023/060959

International Filing Date : 2023.01.20

Dummy Prototypical Networks For Few-Shot Open-Set Keyword Spotting

International Application No. : PCT/US2023/060938

PEER REVIEW EXPERIENCE

The 18th European Conference on Computer Vision (ECCV 2024) 1 paper. Oct. 2024

The Forty-first International Conference on Machine Learning (ICML 2024) 6 papers. Jul. 2024

37th Conference on Neural Information Processing Systems (NeurIPS 2023) 3 papers. Dec. 2023

International Conference on Computer Vision 2023 (ICCV 2023) 5 papers. Oct. 2023

The 16th Asian Conference on Computer Vision (ACCV 2022) 2 papers. Aug. 2022

The IEEE Transactions on Circuits and Systems for Video Technology. Nov. 2021

TECHNICAL SKILLS

- **Languages:** Python, C++, Bash/Shell Scripting
- **Deep Learning Frameworks:** PyTorch, JAX, TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate)
- **Research Areas:** Vision-Language-Action (VLA) Models, Reinforcement Learning (RL), Large Language Models (LLM)
- **Development & Infrastructure:** Linux, Docker, Git, Distributed Training (DDP, FSDP)

INDUSTRY PROJECT

Development of multimodal sensor-based intelligent systems for outdoor surveillance robots

Mar. 2019 - Jun. 2020, Jan. 2021 - Dec. 2021

LG electronics, ETRI, KIRO, **Seoul National University** and Ministry of Science and ICT

Developer

- **Description :** Developed object detection algorithm for outdoor surveillance robots using multi-modal sensors.

FUNDING

Samsung Electronics Ph.D. Student Sponsorship Program

Sep. 2021 - Feb. 2024

HONOUR & AWARDS

The best paper award in the 3rd Workshop on Real-World Surveillance: Applications and Challenges at WACV2023, 1,000 USD Jan. 2023

BK21 Graduate School Innovation Project Colloquium Outstanding Graduate Student Award
500,000 Won (About 420 USD) Jul. 2021

Qualcomm Innovation Fellowship Korea 2020 Final Session, Finalist Dec. 2020

Samsung Electronics - Seoul National University, Industry-Academic Cooperation Research Award
1,500,000 Won (About 1,300 USD) Jul. 2020

Special Award, Yonsei Programming Contest 2018 Spring Semester 2018

Honours, tuition support scholarship, Yonsei University Spring Semester 2016

LANGUAGES

Korean: Native

English: Full Professional Proficiency / Bilingual Proficiency

Capable of writing research papers and conducting high-level technical discussions at a native level.
(Demonstrated through 10+ conference publications).